

```
{
    output_b( LEFT);    // Turn Left
}
    if(le==0 && re==1) // left sensor detects white,
right sensor detects black, so right turn ahead
{
    output_b( RIGHT);    //turn right
}
    if(le==1 && re==1)    // Both sensors detect
black, the robot is perpendicular to the black line
{
    output_b( REV);      // Go back
}
}
}
```

Every line of code is explained alongside in the form of comment. You can have more sensors and more cases to make your code efficient. But to start with programming, the code above would be sufficient.

Probably, coding is the most important part of making robot important because once you have made your robot, you can't just keep on changing the circuit or structure of the robot, but you can change the program as many times you want. So, once the robot is constructed, the only way to improve its efficiency is to improvise on the code.

9.1.1 Programming

Having written the code, the next step is to feed this code into the microcontroller. For this purpose you require two things:

- 1.Hardware Interface
- 2.Software Interface

We have written our code in C++, but the problem that microcontroller understands only binary language still remains. So we need software which converts this code into binary language. Such software is called a 'compiler'.

Now, the choice for your software interface depends on what type of hardware interface you choose. A hardware interface is essentially a 'programmer' which connects from one end to the port of your computer, and is connected to the microcontroller at the other end. It is essentially a physical channel through which your code 'flows' from your computer to the